View Review

Paper ID

657

Paper Title

Autonomous skin cancer categorisation using modern deep learning techniques: an analysis

REVIEW QUESTIONS

1. Comments to Authors

The given paper, Autonomous skin cancer categorization using modern deep learning techniques: an analysis, written by Sivakumar R, address the possibility of using deep learning
approaches to classify a skin lesion, namely, the ResNet deep learning model. The paper explains the dire situation when skin cancer is hard to diagnose and suggests a method of skin cancer diagnosis with the help of deep learning. The paper is properly organized, and deals with problem, related work, methodology, results, and conclusion. The methodology section explains how Convolutional Neural Networks (CNNs) and transfer learning, data augmentation, to make comparable improvements in the accuracy of diagnosis were used. The study brings to focus the need of big, well annotated databases in developing robust and generalizable models.
Nonetheless, the paper has some major areas that need correction. There is a reverse terminology use as the title is about skin cancer categorization but the body talks about the classification of healthy and benign outcomes. Such imprecision in the medical language is confusing because benign lesions are not cancerous, but they are not strictly speaking a part of health. The data itself is described not very carefully, as well, pulling a Kaggle URL as a source and not providing information about the exact types of benign lesions noted.
The section that is devoted to the results includes several severe inconsistencies. The text characterizing ResNet101V2 confusion matrix (Fig. 3) lists 20 False Negatives, however the
image indicates such perfect classification with no off-diagonal numbers. This does not agree with the sensitivity of 0.75 of the same models reported. A sharp flaw is how the reported F1-
score of ResNet101V2 of 1.71. in Table 2 is mathematically infeasible because F1-scores cannot be greater than 1.0. Moreover, ROC curves with ResNet50 and ResNet101V2 seem to be incorrectly labeled and have contradicting information in pictures and description.
Lastly, the paper's clumsy language and several grammatical faults hurt it. Additionally, there is a possible error in the conclusion that refers to "real-time farming applications" rather than a medical context. The manuscript needs to be significantly revised in light of these basic problems in order to be published and